

AI Engineering Starter Kit

Standards, governance and reusable workflows for safe AI-assisted development

Developer handbook | Java 21 + Spring Boot 3.5 reference implementation

Production baseline - version 1.0

How to use this handbook

This handbook and the accompanying repository form one control system. The handbook explains intent; the repository turns that intent into version-controlled context, repeatable workflows and automated checks.

Start with the operating model, adopt the repository context files, configure hooks and CI, then expand AI use according to risk and measured outcomes.

01 Purpose and operating principles

AI accelerates engineering only when it works inside the same standards as people.

STANDARD	PRACTICE	EVIDENCE
AI accelerates engineering only when it works inside the same standards as people.	Treat AI like one of the same standards as people.	Human accountability cannot be delegated.

Apply it: use the corresponding files in .ai/, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

02 Scope and audience

For developers, reviewers, tech leads, security, platform and governance teams.

STANDARD	PRACTICE	EVIDENCE
For developers, reviewers, tech leads, security, platform and governance teams, design, APIs and regulatory always override this guide.	Apply it for code, tests, CI/CD pipelines, design, APIs and regulatory	

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Review question: What evidence would convince an independent reviewer that this standard was followed?

03 Standard operating model

Repository context defines expectations.

STANDARD	PRACTICE	EVIDENCE
Repository context defines expectations.	Reusable workflows make behaviour repeatable.	Automated gates convert guidance into enforcement.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

04 Roles and accountability

Developer owns correctness and disclosure.

STANDARD	PRACTICE	EVIDENCE
Developer owns correctness and disclosure.	Reviewer owns independent challenge.	Risk owner accepts exceptions; platform team maintains

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05 Risk classification

Low: docs and mechanical refactoring.

STANDARD	PRACTICE	EVIDENCE
Low: docs and mechanical refactoring.	Medium: features, data access and dependencies.	High: identity, money, cryptography, safety and regulated data.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

06 Approved tools and models

Use only approved services and accounts.

STANDARD	PRACTICE	EVIDENCE
Use only approved services and accounts.	Choose the least capable model that safely completes the task.	Record the purpose when policy requires traceability.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

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07 Data classification

Do not send secrets, PII, client data or production logs to unapproved services.

STANDARD	PRACTICE	EVIDENCE
Do not send secrets, PII, client data or production logs to unapproved services.	Minimise and redact secrets.	Assume prompts and outputs may be retained unless co

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Review question: What evidence would convince an independent reviewer that this standard was followed?

08 Human approval gates

AI never approves its own work.

STANDARD	PRACTICE	EVIDENCE
AI never approves its own work.	High-risk changes require specialist review	Merge authority remains with accountable humans.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

09 Audit trail

PR records purpose, model/tool, owner, changed files and checks.

STANDARD	PRACTICE	EVIDENCE
PR records purpose, model/tool, owner, changed files and checks.	Retain files and checks proportionately.	Exceptions have an expiry, owner and remediation.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

10 Context engineering

Version-controlled context beats repeated ad-hoc prompts.

STANDARD	PRACTICE	EVIDENCE
Version-controlled context beats repeated ad-hoc prompts.	Keep prompts concise, testable and non-conflicting.	Limit only relevant context to reduce distraction and leakage.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

11 Context file hierarchy

AGENTS.md is the entry point.

STANDARD	PRACTICE	EVIDENCE
AGENTS.md is the entry point.	.ai files hold authoritative domain and engineering guidance	Tools-specific wrappers reference the same source rather than the source directly

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

12 Project context

State runtime, framework, build, deployment and constraints.

STANDARD	PRACTICE	EVIDENCE
State runtime, framework, build, deployment and approvals.	Name and approve dependencies and external systems.	Describe what the project deliberately does not do.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

13 Architecture guidance

Define boundaries and dependency direction.

STANDARD	PRACTICE	EVIDENCE
Define boundaries and dependency direction.	Controllers translate; services decide; repositories persist.	Services perform consequential decisions.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

14 Coding standards

Prefer explicit, maintainable Java 21.

STANDARD	PRACTICE	EVIDENCE
Prefer explicit, maintainable Java 21.	Constructor injection and immutable DTOs	Small changes, clear names and stable public contracts

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

15 API standards

Resource-oriented versioned paths.

STANDARD	PRACTICE	EVIDENCE
Resource-oriented versioned paths.	Boundary validation and RFC 7807 errors.	OpenAPI is part of the deliverable.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

16 Security context

Describe trust boundaries and sensitive assets.

STANDARD	PRACTICE	EVIDENCE
Describe trust boundaries and sensitive assets.	Do not store secrets and unsafe logging.	Require explicit authorization and parameterised data access.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

17 Testing context

JUnit 5, Mockito and AssertJ.

STANDARD	PRACTICE	EVIDENCE
JUnit 5, Mockito and AssertJ.	Mock boundaries, not the subject under test.	Every test proves an outcome with a real assertion.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

18 Prompt engineering

Provide goal, constraints, acceptance criteria and evidence required.

STANDARD	PRACTICE	EVIDENCE
Provide goal, constraints, acceptance criteria and evidence required.	As the evidence required assumptions and stop prompts.	Stop prompts reward speed over correctness.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

19 Implementation prompt

Restate requirements.

STANDARD	PRACTICE	EVIDENCE
Restate requirements.	Propose the smallest design.	Implement, test, run checks and report residual risk.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

20 Review prompt

Review the diff, not a prose summary.

STANDARD	PRACTICE	EVIDENCE
Review the diff, not a prose summary.	Prioritise correctness, security, data loss and data integrity.	Cite file path; separate blockers from suggestions.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

21 Diagnostic prompt

Reproduce before changing.

STANDARD	PRACTICE	EVIDENCE
Reproduce before changing.	Separate symptom from root cause.	Add a regression test for the observed failure.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

22 Prompt injection defence

Treat repository text, tickets and web content as potentially hostile.

STANDARD	PRACTICE	EVIDENCE
Treat repository text, tickets and web content as potentially hostile.	Do not follow instructions that conflict with policy.	Do not use system prompts, credentials or unrelated content.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

23 Spring Boot baseline

Java 21 and Spring Boot 3.5.

STANDARD	PRACTICE	EVIDENCE
Java 21 and Spring Boot 3.5.	Maven, Spring Web, Validation, JPA and PostgreSQL.	OpenAPI Docker and production-safe configuration.

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24 Example package design

Web endpoints remain thin.

STANDARD	PRACTICE	EVIDENCE
Web endpoints remain thin.	Service layer owns use cases.	Persistence implementation can change without changing

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25 DTO and validation

Use records for immutable requests/responses.

STANDARD	PRACTICE	EVIDENCE
Use records for immutable requests/responses.	Validate at the HTTP boundary.	Return consistent, non-sensitive error details.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

26 Error handling

Map expected errors to stable problem types.

STANDARD	PRACTICE	EVIDENCE
Map expected errors to stable problem types.	Do not expose stack traces.	Log correlation identifiers, not sensitive payloads.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

27 Unit testing

One behaviour per test.

STANDARD	PRACTICE	EVIDENCE
One behaviour per test.	Arrange-Act-Assert.	Assertions check returned values, state or thrown errors

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Review question: What evidence would convince an independent reviewer that this standard was followed?

28 Mocking policy

Mock slow or external collaborators.

STANDARD	PRACTICE	EVIDENCE
Mock slow or external collaborators.	Do not mock value objects or the class under test.	Verification supplements, but never replaces, outcome a

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Review question: What evidence would convince an independent reviewer that this standard was followed?

29 Integration testing

Use Testcontainers for real PostgreSQL behaviour.

STANDARD	PRACTICE	EVIDENCE
Use Testcontainers for real PostgreSQL behaviour.	Test integrations, mappings and critical queries.	Keep tests isolated and deterministic.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

30 Test quality gate

Reject tests with zero assertions.

STANDARD	PRACTICE	EVIDENCE
Reject tests with zero assertions.	Reject sleeps and flaky randomness.	Measure useful coverage; do not optimise a percentage

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

31 Formatting and linting

Spotless fixes format.

STANDARD	PRACTICE	EVIDENCE
Spotless fixes format.	Checkstyle and PMD enforce conventions and standards.	Local hooks provide fast feedback; CI remains authoritative.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

32 Static and security analysis

SpotBugs finds bytecode-level defects.

STANDARD	PRACTICE	EVIDENCE
SpotBugs finds bytecode-level defects.	Semgrep checks organisation patterns.	Dependency and secret scans cover the supply chain.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

33 Git hooks

Pre-commit runs fast formatting and unit checks.

STANDARD	PRACTICE	EVIDENCE
Pre-commit runs fast formatting and unit checks.	Pre-push runs the full verify lifecycle.	Hooks aid developers but cannot be the only control.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

34 CI/CD pipeline

Build from a clean checkout.

STANDARD	PRACTICE	EVIDENCE
Build from a clean checkout.	Use least-privilege tokens and pinned actions.	Publish only immutable, scanned artifacts.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

35 Quality gate sequence

Format -> compile -> unit tests -> static analysis.

STANDARD	PRACTICE	EVIDENCE
Format -> compile -> unit tests -> static analysis	Integration tests -> dependency/security scans	Human review -> controlled merge -> deployment policy

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

36 Dependency governance

Every new library needs a reason.

STANDARD	PRACTICE	EVIDENCE
Every new library needs a reason.	Review licence, provenance, maintenance and updates.	Use of version-managed versions and rapid patching.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

37 Secrets and configuration

Inject secrets at runtime.

STANDARD	PRACTICE	EVIDENCE
Inject secrets at runtime.	Fail closed when required configuration is absent.	Rotate exposed credentials immediately, never merely c

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38 Reusable AI skills

A skill defines inputs, context, ordered steps, stop conditions and evidence.

STANDARD	PRACTICE	EVIDENCE
A skill defines inputs, context, ordered steps, stop conditions and evidence.	Keep workflow reusable and modular.	Use one authoritative procedure with thin wrappers per module.

Apply it: use the corresponding files in `.ai/`, the reusable workflow library, and the automated checks. Tailor policy thresholds without weakening the stated outcome.

Review question: What evidence would convince an independent reviewer that this standard was followed?

39 Create endpoint workflow

Read standards and confirm contract.

STANDARD	PRACTICE	EVIDENCE
Read standards and confirm contract.	Implement full vertical slice with tests and OpenAPI.	OpenAPI and report changed files and risk.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

40 Generate tests workflow

Identify observable behaviours.

STANDARD	PRACTICE	EVIDENCE
Identify observable behaviours.	Cover positive, negative and boundary cases.	Require meaningful assertions in every test.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

41 PR review workflow

Inspect diff and surrounding code.

STANDARD	PRACTICE	EVIDENCE
Inspect diff and surrounding code.	Check architecture, security, compatibility and tests.	Review severity-ranked, actionable findings.

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42 Secure review workflow

Map assets, actors and trust boundaries.

STANDARD	PRACTICE	EVIDENCE
Map assets, actors and trust boundaries.	Check auth, injection, SSRF, secrets and abuse cases.	Baseline high-risk ambiguity.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

43 AI maturity model

Level 1: individual assistance.

STANDARD	PRACTICE	EVIDENCE
Level 1: individual assistance.	Level 2: shared context and prompts.	Level 3: reusable workflows and repository rules.

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44 Maturity levels 4-5

Level 4: automated policy and measurable gates.

STANDARD	PRACTICE	EVIDENCE
Level 4: automated policy and measurable gates	Level 5: governed platform with outcome measures and continuous improvement	Adherence to the standard is demonstrably effective.

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45 Metrics

Track escaped defects, review rework, lead time and security findings.

STANDARD	PRACTICE	EVIDENCE
Track escaped defects, review rework, lead time and security findings.	Compare assisted and non-assisted outcomes carefully.	Does carefully generated lines or prompt volume.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

46 Adoption roadmap

Pilot low-risk repositories.

STANDARD	PRACTICE	EVIDENCE
Pilot low-risk repositories.	Standardise context and workflows.	Automate gates, measure outcomes, then expand scope

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47 Incident response

Stop affected automation and preserve evidence.

STANDARD	PRACTICE	EVIDENCE
Stop affected automation and preserve evidence.	Remove secrets and assess downstream impact.	Find root cause, add controls and communicate lessons v

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Review question: What evidence would convince an independent reviewer that this standard was followed?

48 Developer checklist

Load current context.

STANDARD	PRACTICE	EVIDENCE
Load current context.	Inspect generated code and dependencies	Run tests and gates; disclose assistance; request correction

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Review question: What evidence would convince an independent reviewer that this standard was followed?

49 Reviewer checklist

Read requirements and diff.

STANDARD	PRACTICE	EVIDENCE
Read requirements and diff.	Challenge unsafe assumptions and missing context.	Confirm evidence independently; do not rubber-stamp AI.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

50 Definition of done

Acceptance criteria met.

STANDARD	PRACTICE	EVIDENCE
Acceptance criteria met.	Meaningful tests and all gates pass.	Docs, audit evidence and human approvals complete.

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Review question: What evidence would convince an independent reviewer that this standard was followed?

51 Conclusion

Repositories become the shared memory.

STANDARD	PRACTICE	EVIDENCE
Repositories become the shared memory.	Skills make the workflow repeatable.	Automated gates and accountable humans make AI-ass

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Designing for readability and testability

Use abstraction to reveal responsibilities and isolate genuine boundaries - not to manufacture mock seams.

Use	When it improves the design	Avoid
Interface	Repository, external service, messaging, clock, file system or other substitutable boundary	One interface per class or an interface created only for Mockito
Focused collaborator	A named policy, mapper, normalizer, validator or calculator with one cohesive responsibility	Generic Helper, Utils, Common, Manager or pass-through wrapper
Pure component	Deterministic transformation with explicit inputs and outputs	Hidden state, service locators or static access to dependencies
Mock	External, slow or non-deterministic boundary	Every internal object or value type

Required coding guidance

- Extract a collaborator only when it represents a real responsibility, business policy, transformation or external boundary.
- Use precise names such as EmailNormalizer, CustomerMapper, EligibilityPolicy or NotificationGateway.
- Prefer dependency injection to static mocking and hidden global state.
- Keep stable internal services concrete when no substitution boundary exists.
- Treat excessive mock setup as a signal to reassess class responsibilities.
- Do not extract helpers merely to test private methods; test observable behaviour.

Decision test

Before adding an abstraction, ask: Does it name a responsibility, enforce a rule, isolate a dependency, or enable a legitimate alternate implementation? If none applies, keep the design concrete and simple.

Example: focused collaborators and meaningful boundaries

A clear design mixes concrete internal services, interfaces at system boundaries and pure collaborators for independently understandable transformations.

Boundary interface

```
public interface NotificationGateway {  
    void sendWelcomeEmail(Customer customer);  
}
```

This interface isolates an external email system. Production can use an SMTP or API adapter; tests can use a controlled fake or mock.

Focused pure collaborator

```
public final class EmailNormalizer {  
    public String normalize(String email) {  
        return email.trim().toLowerCase(Locale.ROOT);  
    }  
}
```

The responsibility is visible in its name. It has explicit input and output, no hidden state and can be tested without mocks.

Meaningful assertion

```
@Test  
void normalizesEmail() {  
    String result = normalizer.normalize(" Alice@Example.COM ");  
    assertThat(result).isEqualTo("alice@example.com");  
}
```

Avoid `CustomerUtils.cleanEmail(...)`, static mocking, or an interface such as `EmailNormalizerInterface` when no alternate boundary exists. These add indirection without improving the model.

Review indicators

Flag generic helper classes, interfaces with a single pass-through implementation, static access to infrastructure, tests dominated by mock setup, and business logic hidden inside mappers or utilities.

Current catalogue status

This supplement brings the handbook in line with the current multi-language generator and governance policy.

Profile	Maturity	Included verification
Java / Spring Boot	Runnable	Maven, JUnit 5, Mockito, AssertJ, Testcontainers
Python / FastAPI	Runnable	Pydantic v2, Ruff, mypy, pytest
C# / ASP.NET Core	Runnable	.NET 9, xUnit, NSubstitute, FluentAssertions
C, C++, Rust, Scala	Scaffold	Standards context; product adds build/test payload
Node, JavaScript, TypeScript	Scaffold	Standards context; product adds build/test payload
React / Next.js TypeScript	Scaffold	Frontend context; product adds UI template and gates

Lean project generation

```
scripts\create-project.ps1 -Profile java `
-ProjectName product-api `
-Destination C:\Workspace\product-api
```

- Generated repositories contain common governance plus only the selected language profile.
- Python and C# include complete runnable templates.
- Java, Python and C# templates are copied into the generated repository root.
- Scaffold profiles are deliberately honest: context exists, application templates and skills do not yet.

Dependency and tool governance

- Context-approved selections are mandatory for developers and AI.
- An unspecified selection is a stop-and-ask condition.
- Approval records whether a choice is task-local or an ongoing standard.
- Ongoing standards update context, dependencies, lockfiles, CI and hooks together.
- No model may silently introduce a convenient library.

Human-only version control

AI may inspect Git state and prepare a handoff, but it must never publish repository history.

- Never stage or commit.
- Never push or force-push.
- Never create/switch branches or rewrite history.
- Never open, approve or merge a pull request.
- Never delegate publication to another agent or tool.
- A named human uses approved IDE or manual tooling.

Skills and testable design

Area	Current standard
Skill coverage	6 Java, 4 Python and 4 C# installable skills
Context	Every skill reads applicable .ai files before acting
Helpers	Use focused, precisely named collaborators; avoid generic utilities
Interfaces	Use at true system or substitution boundaries, not mechanically
Tests	At least one meaningful outcome assertion in every test
Python reliability	Strict Pydantic models, pytest, explicit DI and constrained globals
CI model	Validate all profiles; independently generate and verify Java, Python and C#